Library Management System

Developed By: Tunir Dasgupta

Project Type: Console-Based Database Management Application

Project Repository: https://github.com/tunirdasgupta/Library-Management-System

Acknowledgements

I received guidance and encouragement from my Computer Science tutor (Mr. Sudipta Naskar, Assistant teacher of Computer Science at St. Augustine's Day School, Kolkata, email: contact.snaskar@gmail.com) during the design and development of the application.

My computer science tutor helped me publish the project repository on GitHub and prepare associated documents such as Project Overview, Screenshots, Readme etc.

I used LLMs such as ChatGPT and Google Gemini to help with learning and debugging during development. These AI tools helped me fix errors and implement features such as protected password entry, validation of fields (emails, phone number, year etc). I also referenced ChatGPT to help me draft the Readme.md documentation and plan the GitHub project structure.

Introduction

The Library Management System is a console-based Python application designed to help manage common library operations such as book management, member management, issuing and returning books and administrative tasks.

The project helped strengthen my understanding of Python programming, database connectivity, input validation and basic software design principles.

Objective

- To develop a functional library management application using Python
- To implement database operations using MySQL
- To understand concepts such as authentication, validation, and error handling
- To improve problem-solving and programming skills

Key Features

- Administrator Functions
  - Administrator authentication and login
  - Add, update, delete and search books
  - Add, update, delete and search users
  - View all books and users
  - Issue books to members
  - Process book returns
- Member Functions
  - Member authentication and login
  - Search books

- View available books
- Borrow books
- Return books
- View borrowing history
- Create, update, delete and search personal reading notes

Technologies Used

- Python
- MySQL
- MySQL Connector
- SQL

Database Design

The system uses a relational database consisting of the following entities:

Table	Purpose
Admin	Stores administrator credentials
Users	Stores member information
Books	Stores book inventory details
Issued	Tracks issued and returned books
Notes	Stores personal reading notes

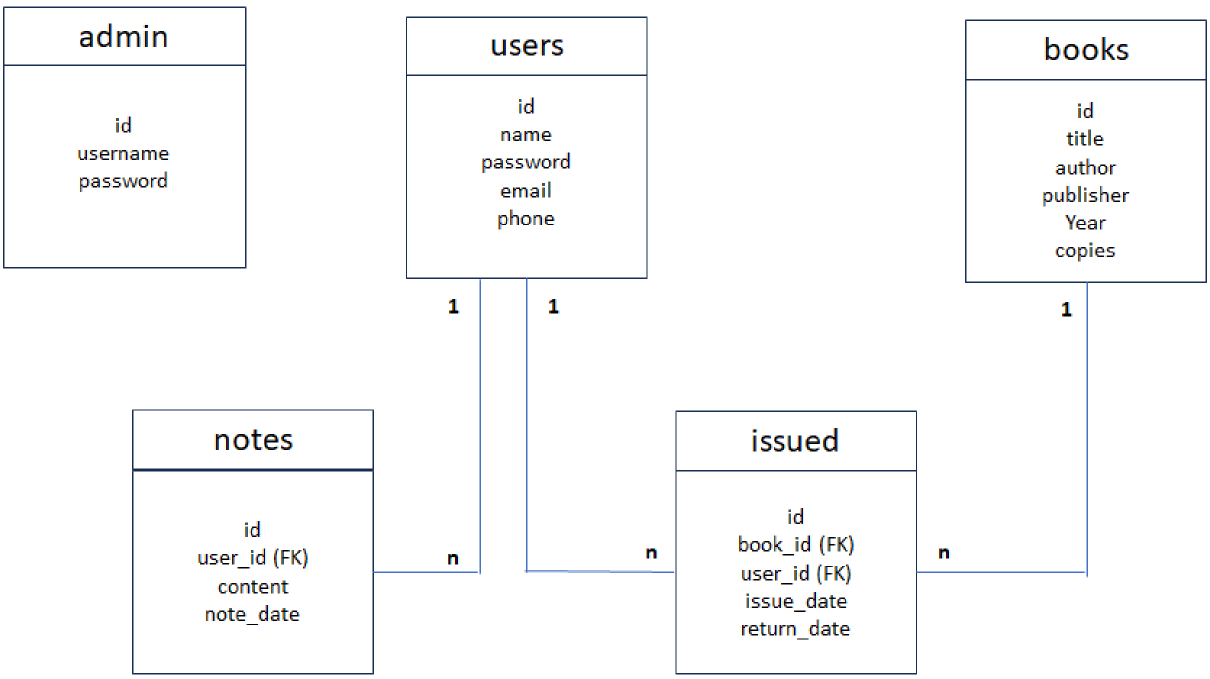

Fig-1: Database Entity Relationship

Security Features

- The project incorporates several security and validation measures:
- Parameterised SQL queries to reduce the risk of SQL injection attacks

- Input validation for user data
- Role-based access control for administrators and members
- Authentication-based access to application features

Future Enhancements

- Graphical user interface
- Password encryption
- Email notifications
- Book recommendation system
- Reporting dashboard

Key Learning Outcomes

- Database design
- Secure coding practices
- Software development lifecycle
- Problem-solving through technology

Conclusion

This project helped me gain practical experience in Python development, database integration, debugging, and documentation. It also improved my understanding of how software projects are structured and maintained using GitHub.

At the same time, I realised that technology is not just about writing code. It is about understanding how people work, how information is organised and how systems can be designed to make processes more efficient. Working on this project strengthened my interest in exploring the connection between technology, society and problem-solving.